Improving Renewable Energy Hosting Capacity in LV Distribution Feeders

Rooftop PV assessment and controlled phase management

A simulation-based capstone project that examines how uneven single-phase solar generation and consumer demand affect low-voltage feeders, and how controlled load transfer can improve voltage balance.



Distribution infrastructure photograph supplied with the project resources. Field context is shown; this is not an installed STS prototype.

Project at a glance	Documented scope
Application	MSEDCL distribution network, Pune
Case study	Sahajanand feeder: 12 LT sections and 179 consumers
PV base	34 PV consumers; 134 kW installed capacity
Implementation	MATLAB/Simulink feeder models and Stateflow STS controller
Key result	Separate steady-state phase-selection tests: VUF 1.5466% to 0.4542%

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Problem statement and project objectives

Why the problem matters

Single-phase rooftop PV and unequal consumer loads create different net power on the R, Y and B phases. Solar output and electricity demand change throughout the day, so a phase allocation that is suitable at one time can become unsuitable later. This can cause voltage rise, voltage unbalance, unequal currents, increased neutral current and reverse power flow.

Distribution utilities commonly rely on load surveys and physical reconnection to rebalance phases. Periodic intervention cannot continuously follow changing generation and demand. The engineering challenge is to identify feeder constraints and evaluate flexible phase management without connecting two supply phases together during transfer.

Objective	Project response
Assess PV impacts	Model phase-wise loading and PV injection; compare voltages, currents and transformer power.
Determine hosting capacity	Establish a baseline and evaluate increasing PV output against defined operating limits. Final maximum capacity remains pending.
Evaluate mitigation	Transfer an eligible single-phase load through an interlocked STS and compare phase-selection outcomes.
Use measured operating data	Analyse May 2026 consumer-meter records and use a feeder snapshot to inform the model.

What hosting capacity means

The maximum additional PV that a specific feeder can accommodate under stated operating conditions before its first applicable technical constraint is reached. Installed PV capacity, tested PV output and hosting capacity are different quantities.

Intended application

Utility feeder assessment, rooftop-PV integration planning, phase-load balancing and power-quality monitoring. MSEDCL is the principal application context in the supplied report.

Source: report Chapters 1-3; updated presentation slides 4-12 and 32.

Engineering approach and key parameters

Two complementary simulation models

STS study: An upstream source, transformers, feeder and unbalanced RL loads are combined with PV current injection and a selected 2 kW load. Stateflow controls its connection to R, Y or B.

Sahajanand study: A three-phase, four-wire feeder is represented by 12 LT sections. The model uses 179 consumers, 34 PV consumers and 134 kW installed PV. Cases V0 and P25 use the same stated load snapshot.

Parameter	Documented value or method
Network frequency	50 Hz
STS network supply chain	22 kV source; 22/11 kV conversion; 11 kV/415 V DTR
Sahajanand DTR	630 kVA, 22 kV/415 V in the updated case-study section
STS LT line model	1.5 km π-section; R = 0.21 ohm; L = 1.5 mH; C = 16.5 nF
Selected load	2 kW; nominal unity power factor; about 8.34 A at 239.6 V
PV representation	PLL-synchronised controlled current sources; PV reactive power set to zero
Snapshot and evaluation	22 May 2026, 14:36; 0.5 s run; settled analysis over 0.2-0.5 s
CT conversion	1000/5 A ratio; multiplier 200 for CT-secondary readings
Load allocation	Total demand shared across 12 LT sections, with phase shares based on measured currents

Workflow

Review literature and field records → construct feeder and load models → establish no-PV baseline → introduce phase-wise PV → measure electrical performance → test phase selection and STS sequencing → identify further validation needs.

Tools: MATLAB, Simulink, Simscape, Stateflow, Excel and CSV processing. OpenDSS is mentioned in the resources, but numerical cross-validation is not provided. Sources: report Chapters 3-5; presentation slides 17-31.

Static transfer switch and voltage balance

Transfer control

The controller receives Phase_Command, Current_Phase, Current_Zero and Transfer_Enable, and produces Gate_R, Gate_Y and Gate_B. A target is latched, the outgoing gate is removed, current zero is awaited, and dead time is applied before the destination gate is enabled.

Stage	Gate state R Y B
Initial connection to Y	0 1 0
Break before make	0 0 0
Destination connection to R	1 0 0

The chart enables transfer after 50 ms and specifies a 10 ms DEAD_TIME state. The illustrated gate waveform shows about 20 ms with all gates off, including sequencing and current-zero waiting. These times describe different parts of the transfer process.

Independent steady-state phase-selection result

Connection of the selected load	Reported sequence VUF
B phase	1.5466%
R phase	0.4542%
Absolute reduction	1.0924 percentage points
Relative reduction	70.63% (calculated from the reported values)

This comparison comes from separate settled simulations. The Y-to-R switching demonstration and the B-versus-R VUF comparison are distinct tests. The preferred phase changes with load and PV conditions; one destination phase is not optimal in every case.

Measurement definition

Sequence VUF = 100 × |V2| / |V1|, where V2 and V1 are negative- and positive-sequence voltage. The report appendix also computes 100 × max |Vphase - Vaverage| / Vaverage, an RMS-magnitude deviation indicator. These measures should be labelled separately.

Sources: report Sections 4.3, 5.1 and 6.2-6.3; updated presentation slides 35-44. Hysteresis and minimum dwell time are discussed, but complete numerical tuning is not supplied.

Sahajanand feeder results

V0 versus P25 at the same stated load snapshot

Metric	No PV V0	PV case P25
PV active power	0.000 kW	38.627 kW
Transformer active power	76.338 kW	42.893 kW
Transformer reactive power	6.563 kvar	6.404 kvar
Transformer apparent power	76.619 kVA	43.368 kVA
Transformer power factor	0.996	0.989
LT1 average voltage	249.11 V	249.34 V
LT6 average voltage	246.92 V	247.69 V
LT12 average voltage	246.00 V	247.00 V
LT1 to LT12 voltage drop	1.25%	0.94%
LT1 unbalance indicator	0.08%	0.10%
LT6 unbalance indicator	0.26%	0.30%
LT12 unbalance indicator	0.37%	0.45%
Section 1 phase A current	80.40 A	33.46 A
Section 12 phase A current	6.70 A	9.10 A
Transformer neutral current	33.60 A	37.41 A
Feeder loss	Not reported	0.237 kW

What the comparison shows

Reported transformer active power falls by 33.445 kW, or 43.8%. LT12 voltage increases by 1.00 V; neutral current increases by 3.81 A. PV therefore reduces upstream demand while changing voltage and phase-current distribution.

The report records no violation of its >254 V, >2% unbalance and negative transformer-power checks in these cases. This does not establish maximum hosting capacity. P25 is a case label: 38.627 kW is 28.8% of 134 kW, not 25%. The appendix uses magnitude deviation for the feeder unbalance indicators. Source: report Section 5.2, printed pp. 62-66.

Measured data and practical context

May 2026 data analysis

The resources describe 487 input rows with 486 valid timestamps and one invalid timestamp. Phase voltage, phase current, power factor, active and reactive power, and frequency were studied across morning, afternoon, evening and night periods.

Reconstructed load statistic	Reported value
Maximum load	145.19 kW; 18 May 2026 at 22:36
Daytime maximum	76.41 kW; 22 May 2026 at 14:36
Minimum / median / mean	5.13 / 67.69 / 63.29 kW

Consumer net-meter measurements and reconstructed feeder demand have different scopes. Raw signed meter kW should not be treated as whole-feeder demand without checking CT scaling, meter polarity and aggregation. Time-of-day variation supports the need for adaptive phase decisions.





Supplied photographs show LT feeder-pillar equipment and phase-cable connections. They provide distribution-network context; the resources do not establish a deployed project controller or hardware trial.

Engineering value

The study brings together site-informed feeder modelling, PV allocation, electrical measurements and interlocked phase transfer. It demonstrates an analysis framework for understanding which operating conditions may limit further rooftop PV integration.

Sources: report Sections 5.3-5.4; updated presentation slides 33 and 41-42; supplied infrastructure photographs.

Team members and individual contributions

Team member and PRN	Documented contribution
Ayush Kumar Dash 1032231558	Performed result analysis and interpretation of measured data in MATLAB. Designed Simulink electrical-distribution models and implemented the Stateflow-based STS.
Chaitanya Pohankar 1032231015	Processed, analysed and modelled measured feeder data using MATLAB and Simulink to obtain load-variation plots for different periods of the day.
Krishna Hambarde 1032231194	Documented the report and simulation work. Conducted an extensive literature survey to understand previous research and relevant approaches.

Academic and industry guidance

Role	Name
Project guide	Dr. Shantaram Dond
Co-guide	Dr. Vinayak Gaikwad
Industry mentors	Assistant Engineer Atul Dhandore Deputy Engineer Kiran Chothe Additional Engineer Vijay Kulkarni

Skills demonstrated

Power-system modelling; three-phase feeder analysis; rooftop-PV integration; MATLAB data processing; Stateflow control sequencing; interpretation of voltage and current measurements; technical literature review; and engineering documentation.

Contribution attribution

Individual responsibilities above are paraphrased from the supplied updated presentation, slide 58. Technical outcomes are presented as team results; the materials do not assign ownership percentages or a team-lead designation.

Sources: report cover and acknowledgements; updated presentation slides 1 and 58; updated poster header.

Project outcomes and next steps

Completed outcomes

Distribution-network and 12-LT Sahajanand feeder models were developed in simulation. A selected 2 kW load-transfer sequence was demonstrated. Separate steady-state tests showed a 70.6% reduction in sequence VUF with suitable phase selection. The feeder V0/P25 comparison and measured-data analysis provide a basis for further PV assessment.

Current limits of the evidence

All controller results are simulation-based. No fabricated STS, PCB, embedded firmware or prototype measurements are documented. PV uses simplified current injection. Machine-learning phase selection is exploratory; no validated model is supplied. No final maximum hosting capacity or quantified hosting-capacity increase is established.

Priorities for further work

Priority	Required next step
Model consistency	Reconcile nominal-voltage descriptions, transformer configurations, solver settings and load/PV power balance.
Measurement consistency	Keep sequence VUF separate from magnitude deviation; verify CT scaling and meter sign conventions.
Hosting-capacity sweep	Increase PV across locations, phases and daily profiles; apply consistent voltage, thermal, neutral-current and export criteria.
Control validation	Use matched settled before/after cases; verify dead time, current-zero detection, interlocks and transfer interruption.
Practical development	Build a protected laboratory prototype, followed by hardware-in-the-loop testing and validated phase-selection logic.

Research status

The report states that “Voltage Unbalance Analysis on LV Distribution Feeders with Rooftop PV” was submitted/applied for IEEE PUNECON 2026. Acceptance and publication are not evidenced in the supplied material. No patent outcome is documented.

Source record

Prepared from Final End term Capstone report.pdf, Updated Capstone PPT(1).pptx, Capstone_Poster_Updated(1).pptx and the supplied photographs. Report references include IEC 61000-4-30, IEC 61000-3-13, IEEE 1547 and IEC 61727 as technical references, not evidence of certification. Conflicting voltage limits (243.8 V and 254 V) require reconciliation before a formal capacity claim.